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NEW PATTERN (2026-27)
BLUE PRINT of CHEMISTRY Q' Paper

CHAPTER (Total Marks)	OMQ(1M) 9×1= 9	VSAQ(2M) 14×2 = 28	SAQ(4M) 8×4 = 32	LAQ(8M) 2×8 = 16	Total 85 M
1. Solutions **(9 M)**	1	2+2	4	–	**9**
2. Electrochemistry **(13 M)**	1	2+2	4+4	–	**13**
3*. Chemical Kinetics **(15 M)**	1	2	4	8	**15**
4. The d&f -Block Elements **(8 M)**	--	2+2	4	–	**8**
5*. Coordination Compounds **(15 M)**	1	2	4	8	**15**
6. Haloalkanes & Haloarenes **(7 M)**	1	2	4	–	**7**
7. Alcohols, Phenols & Ethers **(11 M)**	1	2	4+4	–	**11**
8*. Aldehydes, Ketones & Carboxylic acids **(15 M)**	1	2	4	8	**15**
9. Amines **(7 M)**	1	2	4	–	**7**
10. Biomolecules **(9 M)**	1	2+2	4	–	**9**
Section wise Marks/ Total No.of Q's	**9M/9Q**	**28 M/14Q**	**32M/12Q**	**16M/3Q**	**85 /38Q**

☛ * indicates LAQ Chapters.

'*' *The Blue print is prepared according to the 'Model Question Paper' issued by B.I.E.*

☞ *Please, note that, at times, the public question paper gets slight deviation from the above given Blue Print.*



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☞ This chapter index helps faculty to revise lessons chapterwise.

☞ To get **'Synopsis & Chem Beats'** of respective chapters please scan the above **'QR Code'** through **Google Lens**

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☞ The Chapters are **Renumbered** as per VSAQ, SAQ, LAQ division.

30 DAYS ACTION PLAN

DAY	CHAPTER	REMARKS
Day-1	1) SOLUTIONS (1M + 2M +2M + 4M = 9 M)	OMQ + VSAQ
Day-2	SOLUTIONS	VSAQ + SAQ
Day-3	SOLUTIONS	SAQ + PROBLEMS
Day-4	2) ELECTRO CHEMISTRY (1M + 2M +2M + 4M + 4M = 13 M)	OMQ + VSAQ
Day-5	ELECTRO CHEMISTRY	SAQ
Day-6	ELECTRO CHEMISTRY	SAQ + PROBLEMS
Day-7	3) CHEMICAL KINETICS(1M + 2M + 4M + 8M = 15 M)	LAQ
Day-8	CHEMICAL KINETICS	SAQ
Day-9	CHEMICAL KINETICS	VSAQ + OMQ
Day-10	CHEMICAL KINETICS **Total Marks in this unit : 37 Marks**	REVISION & SLIP TEST -1
Day-11	4) d&f BLOCK ELEMENTS (1M + 2M + 2M + 4M = 9 M)	OMQ + VSAQ
Day-12	d&f BLOCK ELEMENTS	VSAQ + SAQ
Day-13	d&f BLOCK ELEMENTS	REVISION
Day-14	5) CO-ORDINATION COMPOUNDS(1M + 2M + 4M + 8M = 15 M)	LAQ
Day-15	CO-ORDINATION COMPOUNDS	SAQ
Day-16	CO-ORDINATION COMPOUNDS	VSAQ + OMQ
Day-17	CO-ORDINATION COMPOUNDS	REVISION
Day-18	10) BIOMOLECULES (1M + 2M + 2M + 4M = 9 M)	OMQ + VSAQ + SAQ
Day-19	BIOMOLECULES **Total Marks in this unit : 33 Marks**	REVISION & SLIP TEST -2
Day-20	6) HALOALKANES & HALOARENES (1M + 2M + 4M = 7 M)	OMQ + VSAQ
Day-21	HALOALKANES & HALOARENES	SAQ Total: 36 Marks
Day-22	7) ALCOHOLS, PHENOLS & ETHERS(1M + 2M + 4M + 4M = 11 M)	OMQ + VSAQ
Day-23	ALCOHOLS, PHENOLS & ETHERS	SAQ
Day-24	ALCOHOLS, PHENOLS & ETHERS	REVISION
Day-25	8) ALDEHYDES, KETONES & CA (1M + 2M + 4M + 8M = 15 M)	LAQ
Day-26	ALDEHYDES, KETONES & CARBOXYLIC ACIDS	SAQ
Day-27	ALDEHYDES, KETONES & CARBOXYLIC ACIDS	VSAQ + OMQ
Day-28	ALDEHYDES, KETONES & CARBOXYLIC ACIDS	REVISION
Day-29	9) AMINES (1M + 2M + 4M = 7 M)	OMQ + VSAQ
Day-30	AMINES **Total Marks in this unit : 40 Marks**	SAQ REVISION & SLIP TEST -3

☞ **30 Days** x 1 Hour Reading Time (or) **3 Days** x 10 Hours Reading Time

★ 8 MARKS LAQ HIT LIST

TOP 9

Ans-Page

★★★ CHEMICAL KINETICS [8 MARKS] ★★★

LAQ 1. Give a detailed account of the collision theory of reaction rates of bimolecular gaseous reactions. [☞P 14(1)]

LAQ 2. a)Write the differences between order & molecularity. [☞P 15(2)]
b) Describe the salient features of collision theory.

LAQ 3. Explain the following terms with suitable examples. [☞P 16(3)]
a)Average rate of a reaction b) Slow and fast reactions
c)Order of a reaction d)Molecularity of a reaction
e)Activation energy of a reaction

★★★ COORDINATION COMPOUNDS [8 MARKS] ★★★

LAQ 4. a)Write four postulates of Werner's theory of coordination compounds. [☞P 17(4)]
b)Explain cis-trans isomerism in $[Pt(NH_3)_2Cl_2]$ complex.

LAQ 5. Explain the applications of coordination compounds in different fields. [☞P 18(5)]

LAQ 6. Explain different types of isomerism exhibited by coordination compounds, giving suitable examples. [☞P 19(6)]

★★★ ALDEHYDES, KETONES & CARBOXYLIC ACIDS [8 MARKS] ★★★

LAQ 7. Explain the following reactions [☞P 22(8)]
a) Aldol reaction b) Esterification
c) H.V.Z reaction d) Bromination of Benzoic acid

LAQ 8. Describe the following: [☞P 23(9)]
i)Acetylation ii) Cannizzaro reaction
iii)Cross aldol condensation iv) Decarboxylation

LAQ 9. How do you distinguish the following pairs of compounds? [☞P 24(10)]
i)Propanal and propanone ii) Acetophenone and benzophenone
iii)Phenol and benzoic acid iv) Pentan-2-one and Pentan-3-one

★ 4 MARKS SAQ HIT LIST

TOP 40

		Ans-Page
	★★★ SOLUTIONS [4 MARKS] ★★★	
SAQ 1.	What is meant by positive deviations from Raoult's law and how is the sign of Δ_{mix} H related to positive deviation from Raoult's law ?	[☞P 27(12)]
SAQ 2.	What is relative lowering of vapour pressure? How is it useful to determine the molar mass of a solute?	[☞P 28(14)]
SAQ 3.	Define mass percentage, volume percentage and mass to volume percentage solutions.	[☞P 29(16)]
SAQ 4.	A solution of glucose in water is labeled as 10% w/w. What would be the molality of the solution ?	[☞P 29(18)]
SAQ 5.	Calculate the mass of a non-volatile solute (molar mass 40g mol^{-1}) which should be dissolved in 114g Octane to reduce its vapour pressure to 80%.	[☞P 28(15)]
	★★★ ELECTRO CHEMISTRY [4+4 MARKS] ★★★	
SAQ 6.	What are Galvanic cells? Explain the working of Galvanic cell with a neat sketch taking Daniel cell as example.	[☞P 30(19)]
SAQ 7.	State and explain Kohlrausch's law of independent migration of ions.	[☞P 30(20)]
SAQ 8.	Give the construction and working of a standard hydrogen electrode with a neat diagram.	[☞P 31(21)]
SAQ 9.	What are the products obtained at the cathode and the anode during the electrolysis of the following when platinum electrodes are used in the electrolysis (a) Molten KCl (b) Aq. $CuSO_4$ solution (c) Aq.K_2SO_4 solution	[☞P 31(22)]
SAQ 10.	State and explain Nernst equation with the help of a metallic electrode and a non-metallic electrode.	[☞P 32(23)]
SAQ 11.	A solution of $CuSO_4$ is electrolysed for 10 minutes with a current of 1.5 amperes. What is the mass of copper deposited at the cathode?	[☞P 33(25)]
	★★★ CHEMICAL KINETICS [4 MARKS] ★★★	
SAQ 12.	What is 'molecularity' of a reaction? How is it different from the 'order' of a reaction? Name one bimolecular and one trimolecular gaseous reactions.	[☞P 34(28)]
SAQ 13.	Derive an integrated rate equation for a first order reaction.	[☞P 35(30)]
SAQ 14.	What is half-life ($t_{1/2}$) of a reaction? Derive the equations for the 'half-life' value of zero and first order reactions.	[☞P 35(31)]
SAQ 15.	What is Arrhenius equation? Derive an equation which describes the effect of rise of temperature (T) on the rate constant (k) of a reaction.	[☞P 36(33)]
	★★★ d&f BLOCK ELEMENTS[4 MARKS] ★★★	
SAQ 16.	Write the characteristics properties of transition elements.	[☞P 37(34)]
SAQ 17.	What is lanthanoid contraction? What are the consequences of lanthanoid contraction?	[☞P 37(35)]
SAQ 18.	Write the characteristics of interstitial compounds.	[☞P 38(37)]
SAQ 19.	Why do the transition metal ions exhibit characteristic colours in aqueous solution. Explain giving examples.	[☞P 38(38)]

	★★ COORDINATION COMPOUNDS[4 MARKS] ★★	
SAQ 20.	Explain Werner's theory of coordination compounds with suitable examples.	[☞P 39(40)]
SAQ 21.	Explain geometrical isomerism in coordination compounds giving suitable examples.	[☞P 39(41)]
SAQ 22.	What are homoleptic and heteroleptic complexes? Give one example for each.	[☞P 40(42)]
SAQ 23.	Using IUPAC norms write the systematic names of the following: (i)$[Co(NH_3)_6]Cl_3$ (ii)$[Pt(NH_3)_2Cl(NH_2\ CH_3)]Cl$ (iii) $[Ti(H_2O)_6]^{3+}$ (iv) $[NiCl_4]^{2-}$	[☞P 40(45)]
	★★ HALOALKANES & HALOARENES[4 MARKS] ★★	
SAQ 24.	Explain the mechanism of S_N1 with example.	[☞P 41(46)]
SAQ 25.	Write the differences between S_N1 and S_N2 reactions.	[☞P 42(49)]
SAQ 26.	Give the IUPAC names of the following compounds. (i) CH_3–CH(Cl)–CH(I)CH_3 (ii) $ClCH_2$.CH=CH–CH_2Br (iii) $(CCl_3)_3CCl$ (iv) $CH_3C(p\text{–}Cl\text{–}C_6H_4)_2$–CH(Br)$CH_3$	[☞P 43(50)]
SAQ 27.	Write the structures of the following organic halides. (i)1-bromo-4-sec-butyl-2 methylbenzene (ii)2-chloro-1-phenylbutane (iii)p-bromochlorobenzene (iv)4-t-butyl-3-iodoheptane	[☞P 43(51)]
	★★★ ALCOHOLS, PHENOLS & ETHERS[4+4 MARKS] ★★★	
SAQ 28.	Give the equations for the preparation of phenol from Cumene.	[☞P 44(52)]
SAQ 29.	Write the products formed by the reduction and oxidation of phenol.	[☞P 44(53)]
SAQ 30.	Explain the acidic nature of phenols and compare with that of alcohols.	[☞P 44(54)]
SAQ 31.	While separating a mixture of ortho and para nitrophenols by steam distillation, name the isomer which will be steam volatile. Give reason.	[☞P 45(55)]
	★★ ALDEHYDES, KETONES & CARBOXYLIC ACIDS[4 MARKS] ★★	
SAQ 32.	What is Tollen's reagent? Explain its reaction with Aldehydes.	[☞P 47(59)]
SAQ 33.	Write the oxidation products of acetaldehyde, acetone and acetophenone.	[☞P 47(60)]
SAQ 34.	Describe the following: i) Crossed aldol condensation ii) Decarboxylation	[☞P 48(63)]
	★★★ AMINES[4 MARKS] ★★★	
SAQ 35.	Explain i)Sandmeyer reaction ii) Gattermann reaction	[☞P 49(64)]
SAQ 36.	Write the reactions of (i) aromatic and (ii) aliphatic primary amines with nitrous acid, highlighting the differences.	[☞P 50(65)]
SAQ 37.	Write the equations involved in the reaction of Nitrous acid with ethylamine and aniline.	[☞P 50(66)]
	★★★ BIOMOLECULES[4 MARKS] ★★★	
SAQ 38.	Give the dietary sources of the following vitamins and name the diseases caused by their deficiency (a) A (b) D (c) E and (d) K	[☞P 52(69)]
SAQ 39.	What are hormones? Give one example for each. (i) steroid hormones (ii) Polypeptide hormones and (iii) amino acid derivatives.	[☞P 52(70)]
SAQ 40.	Write notes on vitamins.	[☞P 53(71)]

★ 2 MARKS VSAQ HIT LIST

TOP 50+

★★★ SOLUTIONS [2+2 MARKS] ★★★

		Ans-Page
VSAQ 1.	Define molarity	[☞P 56(75)]
VSAQ 2.	Define molality.	[☞P 56(76)]
VSAQ 3.	Define mole fraction.	[☞P 56(77)]
VSAQ 4.	State Henry's law.	[☞P 56(78)]
VSAQ 5.	Define osmotic pressure.	[☞P 56(79)]
VSAQ 6.	What are isotonic solutions?	[☞P 56(80)]
VSAQ 7.	What is Cryoscopic constant?	[☞P 57(82)]
VSAQ 8.	Calculate the molarity of a solution containing 5 g of NaOH in 450 ml solution.	[☞P 57(84)]
VSAQ 9.	Calculate molality of 2.5 g of ethanoic acid (CH_3COOH) in 75g of benzene.	[☞P 57(85)]

★★★ ELECTRO CHEMISTRY [2+2 MARKS] ★★★

VSAQ 10.	State Faraday's first law of electrolysis.	[☞P 58(86)]
VSAQ 11.	State Faraday's second law of electrolysis.	[☞P 58(87)]
VSAQ 12.	What is a fuel cell? How is it different from a conventional galvanic cell?	[☞P 58(88)]
VSAQ 13.	What is metallic corrosion? Give one example.	[☞P 58(89)]
VSAQ 14.	Give the electro-chemical reaction that represents the corrosion or rusting of iron.	[☞P 58(90)]
VSAQ 15.	Define molar conductivity and how is it related to conductivity (k)?	[☞P 59(91)]
VSAQ 16.	What is standard hydrogen electrode?	[☞P 59(94)]

★★ CHEMICAL KINETICS [2 MARKS] ★★

VSAQ 17.	What is a pseudo first order reaction? Give one example.	[☞P 60(96)]
VSAQ 18.	Give two examples for gaseous first order reactions.	[☞P 60(97)]
VSAQ 19.	Give two examples for zero Order reactions.	[☞P 60(98)]
VSAQ 20.	Define molecularity of a reaction with an example.	[☞P 60(99)]
VSAQ 21.	Calculate the overall order of a reaction which has the rate expression (a) Rate = $k[A]^{1/2}[B]^{3/2}$ (b) Rate = $k[A]^{3/2}[B]^{-1}$	[☞P 61(105)]

★★★ The d&f BLOCK ELEMENTS [2+2 MARKS] ★★★

VSAQ 22.	What are transition elements? Give examples.	[☞P 62(107)]
VSAQ 23.	Why Zn^{2+} is diamagnetic whereas Mn^{2+} is paramagnetic?	[☞P 62(108)]
VSAQ 24.	Calculate the 'spin only' magnetic moment of $Fe^{2+}_{(aq)}$ ion.	[☞P 62(109)]
VSAQ 25.	Aqueous Cu^{2+} ions are blue in colour, where as Aqueous Zn^{2+} ions are colourless. Why?	[☞P 62(110)]
VSAQ 26.	What is an alloy? Give example.	[☞P 62(111)]
VSAQ 27.	What is mischmetal? Give its composition and uses.	[☞P 63(112)]
VSAQ 28.	What is actinoid contraction?	[☞P 63(113)]

★★ COORDINATION COMPOUNDS [2 MARKS] ★★

VSAQ 29.	What are coordination compounds? Give two examples.	[☞P 64(117)]
VSAQ 30.	What is a double salt? Give example.	[☞P 64(118)]
VSAQ 31.	What is a chelate ligand ? Give example.	[☞P 64(120)]
VSAQ 32.	What is an ambidentate ligand? Give example.	[☞P 64(121)]

★★ HALOALKANES & HALOARENES [2 MARKS] ★★

VSAQ 33.	What are Enantiomers?	[☞P 66(127)]
VSAQ 34.	What are ambident nucleophiles?	[☞P 66(129)]
VSAQ 35.	What is the stereochemical outcome of S_N1 and S_N2 reactions?	[☞P 66(130)]
VSAQ 36.	Write the structure of the following compounds (i) 2-chloro-3-methylpentane (ii)1-bromo-4-sec-butyl-2-methylbenzene	[☞P 66(131)]
VSAQ 37.	Write the isomers of the compound having molecule formula C_4H_9Br?	[☞P 66(132)]

★★★ ALCOHOLS, PHENOLS & ETHERS [2 MARKS] ★★★

VSAQ 38.	Give the reagents used for the preparation of phenol from chlorobenzene.	[☞P 68(137)]
VSAQ 39.	Explain why propanol has higher boiling point than that of the hydrocarbon butane.	[☞P 68(138)]
VSAQ 40.	Alcohols are comparatively more soluble in water than hydrocarbons of comparable molecular masses. Explain this fact.	[☞P 68(139)]
VSAQ 41.	Write the structures for the following compounds 1)Ethoxy Ethane 2) Ethoxybutane 3) Phenoxy ethane	[☞P 69(143)]

★★ ALDEHYDES, KETONES & CARBOXYLIC ACIDS [2 MARKS] ★★

VSAQ 42.	List the reagents suitable to reduce carboxylic acid to alcohol.	[☞P 70(147)]
VSAQ 43.	Write the reaction showing α-halogenation of carboxylic acid and give its name.	[☞P 70(148)]
VSAQ 44.	How do you distinguish acetophenone and benzophenone?	[☞P 70(150)]
VSAQ 45.	Compare the acidic strength of acetic acid, Chloroacetic acid, benzoic acid and Phenol.	[☞P 71(1530)]

★★ AMINES [2 MARKS] ★★

VSAQ 46.	Arrange the following bases in increasing order of their basic strength. Aniline, p-nitroaniline and p-toluidine.	[☞P 72(155)]
VSAQ 47.	Write equations for carbylamine reaction of an aliphatic amine.	[☞P 72(156)]
VSAQ 48.	Arrange the following bases in decreasing order of pK_b values $C_2H_5NH_2$, $C_6H_5NHCH_3$, $(C_6H_5)_2NH$ and $C_6H_5NH_2$.	[☞P 72(158)]

★★★ BIOMOLECULES [2+2 MARKS] ★★★

VSAQ 49.	What are essential and non-essential amino acids? Give one example for each.	[☞P 74(165)]
VSAQ 50.	Differentiate between globular and fibrous proteins.	[☞P 74(167)]
VSAQ 51.	What is glycogen? How is it different from starch?	[☞P 74(168)]
VSAQ 52.	What is Zwitter ion? Give an example.	[☞P 74(170)]
VSAQ 53.	Write the names of three types of RNA.	[☞P 75(171)]
VSAQ 54.	Name the different types of carbohydrates on the basis of their hydrolysis. Give one example for each.	[☞P 75(173)]

BBQ - TIPS

LAQ SCORING CHAPTERS

Easy Chapter: CHEMICAL KINETICS
Imp Chapter: COORDINATION COMPOUNDS
Super Scoring Pages (SSP): 14,15,17,22,23

SAQ SCORING CHAPTERS

Easy Chapters: SOLUTIONS, BIOMOLECULES, THE d&f -BLOCK ELEMENTS
Imp Chapters: ELECTRO CHEMISTRY, CHEMICAL KINETICS, COORDINATION COMPOUNDS, ALCOHOLS PHENOLS AND ETHERS
Super Scoring Pages (SSP): 28, 29, 30, 32, 34, 35, 37, 39, 41, 42, 44, 47, 49, 50, 52, 53

VSAQ SCORING CHAPTERS

Easy Chapters: SOLUTIONS, ELECTRO CHEMISTRY, d&f, BIOMOLECULES
Imp Chapters: CHEMICAL KINETICS, COORDINATION COMPOUNDS, ALCOHOLS, PHENOLS
Super Scoring Pages (SSP): 56, 57, 58, 59, 60, 62, 63, 64, 66, 68, 70, 72, 74, 75

DEAR STUDENT

- **Quick Memory Bytes** → (i) **Answer Clarity** (ii) **Easy Memory** .
- Collision theory LAQ ను Cricket shots and Class room Analogy లతో పోల్చుకుంటూ చదివితే అది మీకు Lifelong గుర్తుంటుంది. మీరు ఏ పనిలోనైనా Success సాధించాలంటే 'Hardwork తోపాటు Smartwork కూడా చేయాలి' అనే Life Lessons మీకు గుర్తుంటాయి.
- క్లిష్టమైన Werner's theory ను కూడా 5 నిమిషాలలోనే మర్చిపోకుండా నేర్చడానికే **Quick Memory Bytes.**
- Organic Chemistry లోని Named Reactions ను Easy గా గుర్తుపెట్టుకోవడానికి **Mnemonics.**
- Chapter concept పై Motivation కావాలంటే **Chem beats** మీ కోసం రెడీ! (on QR Code).
- **Rising Stars** కోసం **Level I** Questions, **Shining Stars** కోసం **Level II** Questions (Star Q Plus).

☞ **SINCERE SUGGESTION:** After completing this BulletQ, Shining stars are advised to go through the Notes given by your Lecturers; follow the next Higher Level Material.

RISING STARS
MOST FAVOURITE SECTION

LAQ

SECTION-D

BBQ turns, "I can't to I can".

Small book, Big Marks.



LEARN WITH SAI SIR
@myself_saikumar_dasari

LAQ CHAPTERS

16 MARKS - 12 PAGES

★★★ 1.	CHEMICAL KINETICS [3 Qs- 3 Pages]	14 - 16
★★★ 2.	COORDINATION COMPOUNDS [4 Qs- 5 Pages]	17 - 21
★★★ 3.	ALDEHYDES, KETONES [4 Qs- 4 Pages]	22 - 25

TWO BIRDS AT ONE SHOT!

WOW TIP 1: Answer of LAQ 1 repeats in LAQ 2.
Order & Molecularity in LAQ 2 repeats in LAQ 3.

WOW TIP 2: Answer of LAQ 4 (Werner's theory) comes again in SAQ 40.
Answer of LAQ 6 comes again in SAQ 41.

WOW TIP 3: In the **organic chapter** few named reactions are already learned in Jr. Inter.

YouTube @myself_saikumar_dasari | Instagram @learn_with_sai_sir | Telegram t.me/lernwithsaisir

Super Scoring Pages

★★★SSP★★★

Page 14,15,17,22,23

Total SS Pages : 5

1 CHEMICAL KINETICS

8 Marks

1. **Give a detailed account of the collision theory of reaction rates of bimolecular gaseous reactions.** **[BMP]**[AP 18,24]

A: The **Collision Theory** proposed by **Arrhenius** explains how fast a chemical reaction occurs. It states that a reaction takes place only when the gas molecules collide **effectively.**

Main Postulates of Collision Theory:

1) Gas molecules always move in continuous random motion:

They move in all directions. For a collision to occur, firstly any two molecules must collide.

2) But, every collision does not produce a product:

Many collisions are ineffective. Only a few collisions lead to product formation.

3) Effective collision is necessary for a reaction. **Rate of reaction** depends on the 'number of effective collisions'.

4) For a collision to be effective, two conditions must be satisfied:

(a) Activation Energy (E_a): The colliding molecules must possess E_a.

(b) Proper Orientation (P) : The molecules must collide in the correct direction.

5) Collision frequency (Z) is the number of collisions occurring 'per unit volume per second'.

6) From Arrhenius equation, specific rate constant $k = PZe^{-Ea/RT}$

where **P** = Probability (orientation) factor, **Z** = Collision frequency, E_a = Activation energy

R = Universal gas constant, **T** = Absolute temperature

7) Effect of Temperature: Increasing temperature increases the **rate of reaction.**

8) Ex: Formation of methanol from bromomethane depends upon the **orientation** of reactants.

$$CH_3Br + \bar{O}H \xrightarrow[\text{Improper orientation}]{\text{Low energy}} \text{No product}$$

$$CH_3Br + \bar{O}H \xrightarrow[\text{Proper orientation}]{\text{Activation Energy}} CH_3OH + \bar{B}r$$

DOES EVERY SHOT MAKE A SIXER? — **MEMORY BYTES**

- Firstly, Bat & Ball should connect. (Collide)
- Batsman should hit the Ball hard enough to cross the boundary line. (Activation Energy E_a)
- Batsman should also Hit in the Right direction (very high or low means **catch out!**). (Proper orientation P)

2. **a) Write the differences between order & molecularity.**

b) Describe the salient features of collision theory.

A: **a) Differences between order & molecularity:**

Order	Molecularity
i) Order is the sum of powers of concentrations of reactant terms.	i) Molecularity is the number of reactant molecules in a single elementary step.
ii) It is determined **experimentally.**	ii) It is determined **theoretically.**
iii) It can have **fractional values** also.	iii) It can have **integral values** only.
iv) It can be zero order, first order....	iv) It can be unimolecular,bimolecular....

b) Salient features of collision theory:

Molecular collision theory was proposed by **Arrhenius.**

It explains the **rate of gaseous bimolecular reactions.**

Main postulates :

1) All collisions do not form products, only a few collisions form the products.
2) A reaction takes place only when reactant molecules collide with proper orientation (P).
3) **Threshold energy (E_T)** is the minimum energy that the reactant molecules must possess to form products. Molecules having the threshold energy are called **activated molecules.**
4) Very few molecules possess E_T at ordinary temperatures.
5) **Activation Energy (E_a)** is the minimum excess energy that reactant molecules must acquire to participate in a chemical reaction. **Formula:** $E_a = E_T - E_R$.
6) Only collisions between the activated molecules with proper orientation form products. Hence such collisions are called activated collisions or fruitful collisions or effective collisions.
7) Specific rate constant $k = PZ.e^{-Ea/RT}$

☞ If you want, you can write this **Part B answer in LAQ1 and vice-versa**

CLASS ROOM- COLLISION - CAP — **MEMORY BYTES**

☞ Every class is a collision, and only **effective collisions** give **Brilliant Results.**

- **Collision (Interaction):** Faculty comes to class and students listen. No class → No learning.
- **Activation Energy (Energetic teaching & active learning):** Faculty should teach with **enthusiasm,** and students should learn actively with curiosity and focus.
  Low energy & Passive learning → Poor results.
- **Proper Orientation (Right teaching & Keen listening):** Faculty should teach according to students' standards, difficulty level of topics, and students should listen carefully to the lessons without any distractions.

Good Faculty + Good Students → Good Results

☞ **Lecturer + Student:** Interaction + Enthusiasm + Right direction = Brilliant Results.

3. **Explain the following terms with suitable examples.**
   a) Average rate of a reaction **b) Slow and fast reactions**
   c) Order of a reaction **d)Molecularity of a reaction**
   e) Activation energy of a reaction

A: **a) Average rate of a reaction:**
It is defined as the 'rate of change of concentration' of reactants or products per unit time.

Formula: Average rate $\frac{\Delta x}{\Delta t} = -\frac{\Delta[R]}{\Delta t} = +\frac{\Delta[P]}{\Delta t}$

Ex: $2HI \rightarrow H_2 + I_2$

b) Slow reaction: It takes a long time to complete. It requires high activation energy.
Ex: Rusting of Iron takes many days or months.
Fast reaction: It occurs within a very short time. It requires low activation energy.
Ex: Reactions between an acid and base . $HCl + NaOH \rightarrow NaCl + H_2O$

c) Order of a reaction:
It is the 'sum of powers of the concentration terms' of the reactants in the rate law equation.
Ex: For a reaction of the form $aA + bB \rightarrow$ products. , Rate = $k[A]^m[B]^n$.
Here the order of the above reaction is (m+n).

d) Molecularity of a reaction:
It is the number of reacting molecules taking part in a single elementary step.

- **Uni molecular:** $A \rightarrow$ products.
  Ex: $C_2H_5Cl \rightarrow C_2H_4 + HCl$. Here the molecularity is 1.
- **Bi molecular :** $A + B \rightarrow$ products.
  Ex: $H_2 + I_2 \rightarrow 2HI$. Here the molecularity is 2.

e) Activation Energy (E_a):
It is the minimum energy that reactant molecules must possess for the reaction to occur.
Formula: $E_a = E_T - E_R$. E_T = Threshold energy , E_R = Energy of reactants.
Ex: When a matchstick is struck against the match box, **friction** supplies the activation energy for the match stick to ignite. [☞Few More LAQ are in Page 91]

QUICK MEMORY BYTES

- **Average rate** - How fast the reaction proceeds.
- **Slow reaction** (Days/ months) - Rusting of iron.
- **Fast reaction** (Seconds) - Acid + Base reaction.
- **Order** - Sum of powers in the rate reaction.
- **Molecularity** - No.of molecules colliding.
- **Activation energy** - Minimum energy required to start the reaction.

2 COORDINATION COMPOUNDS

8 Marks

4. **a) Write four postulates of Werner's theory of coordination compounds.**
   b) Explain cis-trans isomerism in $[Pt(NH_3)_2Cl_2]$ complex. [BMP] [AP 20,22,23,24,25]

A: **Werner's theory** explains the structure, bonding, and isomerism of complex compounds.

a) Postulates of Werner's theory:

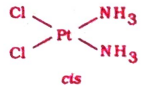
1) The 'central metal ion' in the coordination compounds shows two types of valencies.
   a) Primary valency b) Secondary valency.

2) **Primary Valency:**
   i) It represents the oxidation state (number) of the central metal ion .
   ii) Oxidation state is satisfied (balanced) only by the negative ion ($Cl^-, NO_3^-, ...$)
   iii) These negative ions are ionisable in water.
   iv) These ions lie just outside the coordination sphere.

3) **Secondary Valency:**
   i) It is equal to the co-ordination number of the central metal ion.
   ii) It is satisfied by negative ions (or) neutral molecules-ligands. [$Cl^-, CN^-, NH_3, H_2O, ...$]
   iii) These ligands are not ionisable in water.
   iv) These ions lie inside the coordination sphere in fixed positions and exhibit isomerism.

b) Cis-trans isomerism in $[Pt(NH_3)_2Cl_2]$:

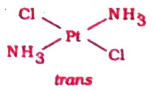
The given complex $[Pt(NH_3)_2Cl_2]$ of the type $[MX_2L_2]$.
The four ligands occupy four corners of a square.



cis

i) **Cis isomer:** Here, two ligands of same type occupy adjacent positions in the coordination sphere.

ii) **Trans isomer:** Here, two ligands of same type occupy opposite positions in the coordination sphere.



trans

QUICK MEMORY BYTE **WERNER - Very Powerful SI**

V- Valency is two types (Primary and Secondary).
P - Primary valency is the oxidation number of the central ion. (Outside - Ionisable)
S - Secondary valency is the coordination number of the central ion.(Inside - Non -ionisable)
I - Isomerism occurs due to identical X ligands .

5. Explain the applications of coordination compounds in different fields.

A: **1) Analysis of Chemicals:** Coordination compounds like EDTA, DMG are used in the estimation of qualitative and quantitative analysis of chemical compounds.

2) Hardness of water : Hardwater contains excessive Ca^{+2}, Mg^{+2} ions.

They are estimated by EDTA.

3) Silver and Gold extraction:

'Complex formation' plays a key role in the extraction of Silver and Gold.

Ex: $AgCl + 2NaCN \rightarrow Na[Ag(CN)_2] + NaCl$

$2Na[Ag(CN)_2] + Zn \rightarrow Na_2[Zn(CN)_4] + 2Ag\downarrow$

4) Metal Purification : 'Metal complexes' play a key role in the purification of metals like Ni, Fe.

Ex: Purification of Ni by Mond's process $[Ni(CO)_4] \xrightarrow{\Delta} Ni + 4CO$

5) Biological systems: Coordination complexes are very important in biological systems.

i) Photosynthesis in plants is due to chlorophyll, a complex of Mg^{+2} ion

ii) Vitamin B_{12} is a complex of Co^{+3} ion.

iii) In Hemoglobin, Heme is the complex of Fe^{+2} ion

6) Catalyst compounds : Some complexes act as catalysts by stabilising the activation energy.

Ex: Wilkinson catalyst is a Rhodium complex $[(Ph_3P)_3RhCl]$ used in hydrogenation of alkenes.

7) Photography: In the process of photo developing when AgBr film is dissolved in hypo solution $[Ag(S_2O_3)_2]^{3-}$ is formed

8) Medical Field: Coordination compounds are used in the medicinal chemistry.

Ex: EDTA is used in the treatment of lead poisoning.

QUICK MEMORY BYTES

A Hard Smart Man Became College Popular Master

A →Analysis (Qualitative and quantitative - EDTA, DM)
H → Hardness water - ions Ca^{+2}, Mg^{+2}
S → Silver & Gold extraction
M → Metal purification
B→ Biological
C→ Catalyst
P→ Photography
M → Medicine

Coordination compounds are simply called **Complexes.**

Telegram t.me/lernwithsaisir
Instagram @learn_with_sai_sir
YouTube @myself_saikumar_dasari

6. Explain different types of isomerism exhibited by coordination compounds, giving suitable examples.

A: Coordination compounds have two types of isomerism: **I) Stereo II) Structural**

I) Stereoisomerism occurs when two complexes have the same molecular formula, but their ligands have different positions in 3D space. This is two types. 1) Geometrical 2) Optical.

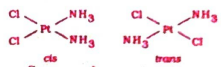
1) Geometrical isomers occur due to different geometric arrangements of the ligands.

i) Geometrical Isomerism in Square planar complexes of the type $[MX_2L_2]$:

Here **4 ligands occupy 4 corner places** of a square around the central ion.

a) cis isomer: Two identical X ligands are arranged adjacent to each other.

b) trans isomer: Two identical X ligands are arranged opposite to each other.



Square planar structure

Ex: cis $[Pt(NH_3)_2Cl_2]$, trans $[Pt(NH_3)_2Cl_2]$

ii) Geometrical Isomerism in Octahedral complexes:

Here **6 ligands occupy 6 corner places** of an octahedral.

a) cis & trans isomer of the type $[MX_2L_4]$:



Octahedral structure

- **cis isomer:** Two X ligands are adjacent to each other.
- **trans isomer:** Two X ligands are opposite to each other.

Ex: cis $[CoCl_2(NH_3)_4]^+$, trans$[CoCl_2(NH_3)_4]^+$

b) fac & mer isomer of the type $[MX_3L_3]$:

- **fac isomer:** Three X ligands occupy one triangular face of the octahedral.
- **mer isomer:** Three X ligands lie along a meridian of the octahedral.

Ex: fac $[Co(NH_3)_3(NO_3)_3]$, mer $[Co(NH_3)_3(NO_3)_3]$.

2) Optical isomers are mirror images that cannot be superimposed on one another(**Enantiomers.**)

i) **d-isomer** rotates the plane polarised light in clock-wise direction (right).

ii) ***l*-isomer** rotates plane polarised light in anticlock-wise direction(left) .

Ex: d-$[Co(en)_3]^{3+}$ and *l*-$[Co(en)_3]^{3+}$

II) Structural isomerism occurs when two complexes have the same molecular formula, but their ligands have different kinds of connections (w.r.to coordination sphere)

1) Ionisation isomers: Here a **ligand inside** the coordination sphere is exchanged with its **counter ion outside** the coordination sphere.

Ex: $[Co(NH_3)_5SO_4]Br$ & $[Co(NH_3)_5Br]SO_4$

2) Hydrate isomers: Here we have different number of water molecules H_2O inside and outside the coordination sphere. **Ex:** $[Cr(H_2O)_6]Cl_3$ & $[Cr(H_2O)_5Cl]Cl_2.H_2O$.

3) Linkage isomers: Here the ambidentate ligands present in coordination sphere differ in the point of attachment. **Ex:** $[Co(NH_3)_5(NO_2)]Cl_2$ & $[Co(NH_3)_5(ONO)]Cl_2$

4) Coordination isomers: Here interchange of ligands between cationic and anionic entities occurs. **Ex:** $[Co(NH_3)_6][Cr(CN)_6]$, $[Co(CN)_6][Cr(NH_3)_6]$.

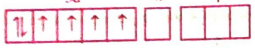
7.1 Discuss the nature of bonding and magnetic behaviour in the following coordination entities on the basis of valence bond theory
(i) $[Fe(CN)_6]^{4-}$ (ii) $[FeF_6]^{3-}$

A: **(i) $[Fe(CN)_6]^{4-}$:** The oxidation state of Fe in this complex is +2.

Fe (Z=26)=[Ar]$3d^6 4s^2$

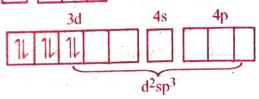


Fe^{+2}(Z=24)=[Ar]$3d^6 4s^0$



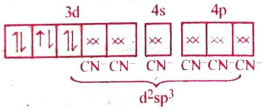
CN^- is a strong ligand.
So forced pairing happens and
hence d^2sp^3 hybridization takes place.

Fe^{+2}



Now six empty d^2sp^3 hybrid orbitals overlap with 6 ligand orbitals (CN^-) to form the complex ion.

$[Fe(CN)_6]^{4-}$



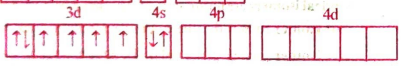
This complex ion is **octahedral**. It is **diamagnetic** as it does not contain any unpaired electron.
Also (n–1)d orbitals are involved in hybridization. So it is a inner orbital or low spin complex ion.

ii) $[FeF_6]^{-3}$: The oxidation state of Fe in this complex is +3.

Fe (Z=26)=[Ar]$3d^6 4s^2$

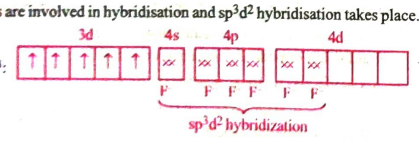


Fe^{+3} (Z=23)= [Ar] $3d^5$ $4s^0$



F^- is a weak ligand. So pairing does not happens .
Here 3d orbitals are not available to take part in bonding.
Hence nd orbitals are involved in hybridisation and sp^3d^2 hybridisation takes place.

$[Fe F_6]^{-3}$:



Because of the presence of five unpaired electrons, the complex is **paramagnetic.**
Also outer d orbitals are involved in hybridization, so it is an **outer orbital** or **high spin complex ion.**

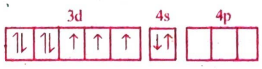
LEARN WITH SAI SIR
@myself_saikumar_dasari



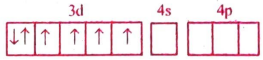
7.2 Discuss the nature of bonding and magnetic behaviour in the following coordination entities on the basis of valence bond theory
(i) $[Co(C_2O_4)_3]^{3-}$ (ii) $[CoF_6]^{3-}$

A: **i) $[Co(C_2O_4)_3]^{3-}$:** The oxidation state of Co in this complex is +3.

Co(Z=27) = [Ar]$3d^7 4s^2$

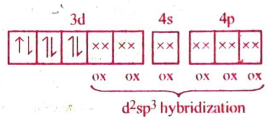


Co^{+3} = [Ar] $3d^6 4s^0$



$C_2O_4^{-2}$ **is a strong ligand** .
So pairing happens and hence d^2sp^3 hybridization takes place.
Now 6 empty d^2sp^3 hybrid orbitals overlap with 6 ligand orbitals ($C_2O_4^{-2}$) to form complex ion.

$[Co(C_2O_4)_3]^{3-}$:



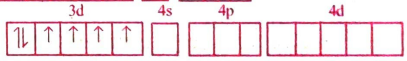
Since, all the electrons are paired, this complex is diamagnetic. It is an inner orbital complex because of the involvement of (n–1)d orbital for bonding.

(ii) $[CoF_6]^{3-}$: The oxidation state of Co in this complex is +3.

Co(Z=27) : [Ar] $3d^7 4s^2$,

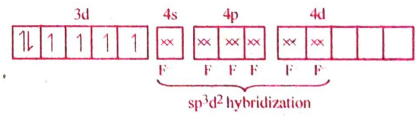


Co^{+3} (Z=24) [Ar] $3d^6 4s^0$



F^- is a weak field ligand. So pairing does not happen
Thus 3d orbitals are not available to take part in bonding.
Hence nd orbitals are involved in hybridisation and sp^3d^2 hybridisation takes place.

$[CoF_6]^{3-}$:



Because of the presence of four unpaired electrons, the complex is **paramagnetic.**
Since outer d orbitals take part in bonding it is an outer orbital complex ion or high spin complex ion.

3 ALDEHYDES, KETONES & CARBOXYLIC ACIDS

8 Marks

8. Explain the following reactions **[BMP]|AP 18,23,24,25|**

a) Aldol reaction **b) Esterification**

c) HVZ reaction **d) Bromination of Benzoic acid**

A: a) Aldol reaction:

Two aldehyde or ketone molecules having 'atleast one α-hydrogen' react together in the presence of a 'dilute alkali catalyst' to form **β-hydroxy aldehyde or β-hydroxy ketone (aldol).**

Ex: $\underset{\text{Acetaldehyde}}{2CH_3CHO} \xrightleftharpoons{\text{dil NaOH}} \underset{\substack{\text{OH}\\ \text{3-hydroxy butanal(aldol)}}}{CH_3-\overset{\overset{OH}{|}}{CH}-CH_2-CHO}$

b) Esterification:

Carboxylic acid reacts with alcohol in the presence of a 'strong acid catalyst' to give **Ester.**

Ex: $\underset{\text{Ethanoicacid}}{CH_3COOH} + \underset{\text{Ethylalcohol}}{C_2H_5OH} \xrightleftharpoons{Conc.H_2SO_4} \underset{\text{(Ester)}}{CH_3COOC_2H_5} + \underset{\text{water}}{H_2O}$

c) HVZ Reaction:

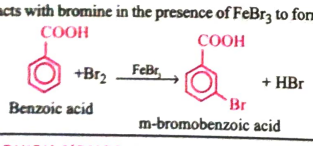
Carboxylic Acid having atleast one α- hydrogen is treated with Halogens (Br (or) Cl) in the presence of red phosphorus undergo Halogenation to form α-Halo Carboxylic Acid.

Ex: $\underset{\text{Carboxylic acid}}{R-CH_2-COOH} \xrightarrow[(ii)\ H_2O]{(i)\ X_2/\text{Redphosphorus}} \underset{\alpha\text{-halo carboxylic acid}}{R-\overset{\overset{X}{|}}{CH}-COOH} + HX$

d) Bromination of Benzoic acid:

Benzoic acid reacts with bromine in the presence of $FeBr_3$ to form **m-bromo benzoic acid.**

Ex:



Benzoic acid — m-bromobenzoic acid

QUICK MEMORY BYTES & MNEMONICS

1. **Aldol:** 2 α Aldehydes → β-hydroxy aldehyde (Already read in Jr. Inter)
2. **Esterification :** Acid + Alcohol = Ester + Water
3. **HVZCAH HCA** [Hello Vizag wala! Come And Have Hyderabad Chicken Biryani at Ameerpet]
4. **Bromination:** Ben - Bro with $FeBr_3$ forms Bromo Benzoic acid.

9. Describe the following: **|AP 19|**

i) Acetylation **ii) Cannizzaro reaction**

iii) Cross aldol condensation **|AP 15,19|** **iv) Decarboxylation**

A: i) Acetylation:

Alcohols when treated with an acyl chloride (acetyl) in the presence of **pyridine** give esters.

Ex: $\underset{\text{(Ethylalcohol)}}{C_2H_5OH} + \underset{\text{(Acetyl chloride)}}{CH_3-\overset{\overset{O}{\|}}{C}-Cl} \xrightarrow{\text{Pyridine}} \underset{\text{(Ethyl acetate)}}{CH_3-\overset{\overset{O}{\|}}{C}-OC_2H_5} + HCl$

ii) Cannizzaro Reaction: Formaldehyde in the presence of conc. NaOH undergo self oxidation and reduction to form Methanol and Sodium Formate.

Ex: $\underset{\text{Formaldehyde}}{2HCHO} \xrightarrow[\text{self.Oxdn\&Redn}]{\text{conc.NaOH}} \underset{\text{Methanol}}{CH_3OH} + \underset{\text{Sodium Formate}}{HCOONa}$

iii) Cross aldol condensation: It is the condensation of two different carbonyl compounds (one of which must have one α-hydrogen) in the presence of a strong base.

$\underset{\substack{\text{Acetaldehyde}\\ \text{(Ethanal)}\\ \text{(A)}}}{CH_3CHO} + \underset{\substack{\text{Propionaldehyde}\\ \text{(Propanal)}\\ \text{(B)}}}{CH_3CH_2CHO} \longrightarrow$

$CH_3-CH=CH-CHO$ (But-2-enal) **(AA)**

$CH_3-CH_2-CH=\underset{\overset{|}{CH_3}}{C}-CHO$ (2-MethylPent-2-enal) **(BB)**

(Self-Aldol Products)

$CH_3-CH=\underset{\overset{|}{CH_3}}{C}-CHO$ (2-Methylbut-2-enal)

$CH_3-CH_2-CH=CH-CHO$ (Pent-2-enal) **(BA)**

(AB) (Cross Aldol Products)

iv) Decarboxylation : Sodium Ethanoate is heated with **Sodalime** to form **Methane.**

Ex: $\underset{\text{Sodium Ethanoate}}{CH_3COONa} + \underset{\text{Soda lime}}{NaOH} \xrightarrow[\Delta]{CaO} \underset{\text{Methane}}{CH_4} + \underset{\text{Sodium carbonate}}{Na_2CO_3}$

QUICK MEMORY BYTES & MNEMONICS

- **Acetylation** → Mixing of acetyl in alcohol gives ester.
- **Cannizzaro Reaction :** [FM Songs Free]

  Formaldehyde formal గా NaOH వద్దకు వస్తే చాలు చల్లటి Methanol & Sodium Formate form అవుతాయి.
- **Cross Aldol** → CROSS - Carbonyl compounds React with each other,

  One must has α-H Strong base (NaOH) → Successful product.
- **Decarboxylation (Removing CO_2) :** DSESM [Dear SESMa నిన్ను]

10. How do you distinguish the following pairs of compounds?

i) Propanal and propanone **ii) Acetophenone and benzophenone**

iii) Phenol and benzoic acid **iv) Pentan-2-one and Pentan-3-one**

A: i) Propanone and Propanal :

Propanone undergoes **idoform reaction**, but Propanal does not undergo idoform reaction.

$$\underset{\text{Propanone}}{CH_3COCH_3} + 3I_2 + 4NaOH \xrightarrow{\Delta} \underset{\text{Idoform}}{CHI_3} + CH_3COONa + 3H_2O + 3NaI$$

ii) Acetophenone and benzophenone:

Acetophenone undergoes **idoform reaction,** but Bezophenone does not undergo it.

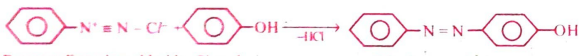
Reason: Acetophenone is a methyl ketone ($C_6H_5COCH_3$)

while benzophenone ($C_6H_5COC_6H_5$) is a diphenyl ketone.

$$\underset{\text{Acetophenone}}{C_6H_5-CO-CH_3} + 3I_2 + 4NaOH \xrightarrow{\Delta} \underset{\text{Idoform}}{CHI_3} + C_6H_5COONa + 3H_2O + 3NaI$$

iii) Phenol and benzoic acid:

- Phenol gives coupling reactions with diazonium compounds, while benzoic acid does not.
- Phenol reacts with benzenediazonium chloride ($C_6H_5N_2Cl$) to give azodyes, but benzoic acid does not react with diazonium salts.
- Phenol gives reddish colour with neutral $FeCl_3$ while benzoic acid gives buff-colour.



Benzene diazonium chloride Phenol P-hydroxy azobenzene (orange dye)

iv) Pentan-2-one and Pentan-3-one:

Pentan-2-one undergoes **idoform reaction**, but pentan-3-one does not give this reaction.

Reason: Pentan-2-one is a methyl ketone.

$$\underset{\text{Pentan-2-one}}{CH_3-\overset{\overset{O}{\|}}{C}-CH_2-CH_2-CH_3} + 3I_2 + 4NaOH \longrightarrow \underset{\text{Idoform}}{CHI_3} + C_3H_7COONa + 3H_2O + 3NaI$$

QUICK MEMORY BYTES

- In (i), (ii), (iv) the first term compounds under go **idoform test** and the second ones do not.

Note: In (i), the terms Propanal and propanone are interchanged in the side heading.

11. Explain the following terms. Give an example of the reaction in each case.

i) Cyanohydrin **ii) Acetal**

iii) Semicarbazone **iv)Aldol**

A: i) Cyanohydrin is carbonyl compound + HCN

Hydrogen cyanide (HCN) adds to aldehydes and ketones to form cyanohydrin.

Ex:

$$\underset{\text{Acetaldehyde}}{HCN + CH_3-\overset{\overset{H}{|}}{C}=O} + \longrightarrow \underset{\text{acetaldehyde cyanohydrin}}{CH_3-\overset{\overset{H}{|}}{\underset{\underset{CN}{|}}{C}}-OH}$$

ii) Acetal is Aldehyde + 2 alcohol molecules

Aldehydes react with alcohols in the presence of dry hydrogen chloride to form acetals.

Ex:

$$\underset{\text{Aldehyde}}{R-\overset{\overset{H}{|}}{C}=O} + \underset{\text{Alcohol}}{R-OH} \xrightarrow[\text{HCl gas}]{\text{dry}} \underset{\text{Semi Acetal}}{R-\overset{\overset{H}{|}}{\underset{\underset{OR}{|}}{C}}-OH} \xrightarrow[\text{dry HCl}]{+ROH} \underset{\text{Acetal}}{R-\overset{\overset{H}{|}}{\underset{\underset{OR}{|}}{C}}-OR} + H_2O$$

iii) Semicarbozone is a carbonyl compound + semicarbazide

Aldehydes and ketones react with semicarbazide ($NH_2NHCONH_2$) to form semicarbazone.

Ex:

$$\underset{\text{Acetaldehyde}}{CH_3-\overset{\overset{H}{|}}{C}=O} + \underset{\text{Semicarbazide}}{H_2N-NH-CONH_2} \longrightarrow \underset{\text{Ethanal semi carbazone}}{CH_3-\overset{\overset{H}{|}}{C}=NNHCONH_2} + H_2O$$

iv) Aldol is a β-hydroxy aldehyde (or) β-hydroxy ketone formed by condensation.

Aldol reaction: Aldehydes and ketones having at least one α-hydrogen undergo a reaction in the presence of dilute alkali as catalyst to form β-hydroxy aldehyde (aldol) or β-hydroxy ketones (ketol) respectively.

Ex:

$$\underset{\text{Acetaldehyde}}{2CH_3CHO} \xrightleftharpoons{\text{dil NaOH}} \underset{\text{3-hydroxy butanal(aldol)}}{CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-CHO}$$

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